

MAT 324 HW01

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Problem 1. Let $\mathcal{A}$ be a collection of closed subsets of $\mathbb{R}^n$ such that $\bigcap_{F \in \mathcal{A}} F = \emptyset$.

- (a) Assume in addition that $\mathcal{A}$ contains a bounded subset F_0 . Show that $\mathcal{A}$ contains a finite subcollection $\mathcal{A}'$ so that $\bigcap_{F \in \mathcal{A}'} F = \emptyset$.
- (b) Give an example showing that the above conclusion need not hold if $\mathcal{A}$ does not contain a bounded set.

Solution. (a) Since the intersection of the sets in $\mathcal{A}$ is empty, we can deduce by de Morgan's law that

$$\bigcup_{F \in \mathcal{A}} \mathbb{R}^n \setminus F = \mathbb{R}^n.$$

Since each F in $\mathcal{A}$ is closed, the complements $\mathbb{R}^n \setminus F$ are open, so $\{\mathbb{R}^n \setminus F : F \in \mathcal{A}\}$ is an open cover of F_0 , which is compact because it is closed and bounded. This means there is a finite subcollection $\{F_1, \dots, F_n\} \subseteq \mathcal{A}$ such that

$$F_0 \subseteq \bigcup_{i=1}^n \mathbb{R}^n \setminus F_i \iff \mathbb{R}^n \setminus F_0 \supseteq \bigcap_{i=1}^n F_i.$$

In particular, the intersection of the F_i is disjoint from F_0 , so $\bigcap_{i=1}^n F_i = \emptyset$; thus, $\{F_1, \dots, F_n\}$ is the desired subcollection of $\mathcal{A}$.

(b) Let $F_n = [n, \infty)$ be a closed subset of $\mathbb{R}$ and let $\mathcal{A} = \{F_n : n \in \mathbb{N}\}$; we show that $\mathcal{A}$ has the desired property. The intersection of all the sets in $\mathcal{A}$ is empty because given $x \in \mathbb{R}$, we can find n such that $x \notin F_n$, namely any $n > x$. However, no finite subcollection of $\mathcal{A}$ has empty intersection because, given $a_1, \dots, a_k \in \mathbb{N}$, we have

$$\bigcap_{i=1}^k F_{a_i} = \bigcap_{i=1}^k [a_i, \infty) = [\max(a_1, \dots, a_k), \infty) \neq \emptyset. \quad \square$$

Problem 2. Suppose $a, b \in \mathbb{R}$ with $a < b$ and $f: [a, b] \rightarrow \mathbb{R}$ is a bounded function.

(a) Let $\varepsilon > 0$. Show that there exists $\delta > 0$ such that

$$L(f, \mathcal{P}) > L(f, [a, b]) - \varepsilon \quad \text{and} \quad U(f, \mathcal{P}) < U(f, [a, b]) + \varepsilon$$

for every partition $\mathcal{P}$ of $[a, b]$ with mesh size $\delta(\mathcal{P}) < \delta$.

(b) Let $\mathcal{P}_n$ be a sequence of partitions of $[a, b]$ such that $\delta(\mathcal{P}_n) \rightarrow 0$. Show that

$$L(f, [a, b]) = \lim_{n \rightarrow \infty} L(f, \mathcal{P}_n) \quad \text{and} \quad U(f, [a, b]) = \lim_{n \rightarrow \infty} U(f, \mathcal{P}_n).$$

Solution. (a) By definition, $L(f, [a, b])$ is the supremum of $L(f, \mathcal{P})$ over all partitions $\mathcal{P}$ of $[a, b]$. Therefore, for every $\varepsilon > 0$, there exists a partition $\mathcal{P}_{\varepsilon/2} = (x_0, x_1, \dots, x_m)$ such that

$$L(f, \mathcal{P}_{\varepsilon/2}) > L(f, [a, b]) - \frac{\varepsilon}{2}.$$

Given an arbitrary partition $\mathcal{P} = (y_0, y_1, \dots, y_n)$ with

$$\delta(\mathcal{P}) := \min_{i=1, \dots, n} y_i - y_{i-1} < \delta := \frac{\varepsilon}{2M(m+1)},$$

we will compare $L(f, \mathcal{P})$ with $L(f, \mathcal{P}')$, where $\mathcal{P}'$ is the common refinement $\mathcal{P} \cup \mathcal{P}_{\varepsilon/2}$ of $\mathcal{P}$ and $\mathcal{P}_{\varepsilon/2}$ (i.e. merge the two partitions to get $\mathcal{P}'$). Keep in mind that $L(f, \mathcal{P}')$ is at least as large as both $L(f, \mathcal{P})$ and $L(f, \mathcal{P}_{\varepsilon/2})$. To do the comparison, observe that at most $m+1$ of the intervals $[y_{i-1}, y_i]$ contain some x_j , since there are only $m+1$ elements in the $\mathcal{P}_{\varepsilon/2}$ list. Clearly, these intervals are the ones that account for the difference between $L(f, \mathcal{P})$ and $L(f, \mathcal{P}')$, since if y_{i-1} and y_i are consecutive in both $\mathcal{P}$ and $\mathcal{P}'$ then the interval $[y_{i-1}, y_i]$ has the same contribution of

$$(y_i - y_{i-1}) \inf_{x \in [y_{i-1}, y_i]} f(x)$$

to both $L(f, \mathcal{P})$ and $L(f, \mathcal{P}')$. Within one of these intervals $[y_{i-1}, y_i]$ that contain some x_j s, we see that the total contribution of this interval to $L(f, \mathcal{P}')$ is bounded above by

$$(y_i - y_{i-1}) \sup_{x \in [y_{i-1}, y_i]} f(x)$$

because the infimum of f over each of the subintervals of $[y_{i-1}, y_i]$ is trivially bounded above by the supremum of f over $[y_{i-1}, y_i]$. Since f is bounded there exists a positive M such that $|f(x)| < M/2$ for all x . The contribution of the interval $[y_{i-1}, y_i]$ to the increase between $L(f, \mathcal{P})$ and $L(f, \mathcal{P}')$ can then be bounded above as

$$(y_i - y_{i-1}) \left(\sup_{x \in [y_{i-1}, y_i]} f(x) - \inf_{x \in [y_{i-1}, y_i]} f(x) \right) < \delta \cdot M.$$

There are up to $m+1$ such intervals $[y_{i-1}, y_i]$, meaning $L(f, \mathcal{P}') - L(f, \mathcal{P})$ can be bounded above by $M(m+1)\delta = \frac{\varepsilon}{2}$. Therefore,

$$L(f, \mathcal{P}) > L(f, \mathcal{P}') - \frac{\varepsilon}{2} \geq L(f, \mathcal{P}_{\varepsilon/2}) - \frac{\varepsilon}{2} > L(f, [a, b]) - \varepsilon,$$

which is what we wanted.

For the upper Riemann integral, we can use the fact that $U(f, [a, b]) = -L(-f, [a, b])$ and $U(f, \mathcal{P}) = -L(-f, \mathcal{P})$ for any partition $\mathcal{P}$, so, given δ such that $\delta(\mathcal{P}) < \delta$ implies $L(-f, \mathcal{P}) > L(-f, [a, b]) - \varepsilon$,

$$U(f, \mathcal{P}) = -L(-f, \mathcal{P}) < -L(-f, [a, b]) - \varepsilon = U(f, [a, b]) - \varepsilon.$$

Since we want one δ to satisfy both the condition on L and on U , we can just take δ to be the minimum of the two δ s obtained for L and U above.

(b) Given $\varepsilon = \frac{1}{m}$ for some positive integer m , part (a) posits the existence of a $\delta_m > 0$ such that a partition $\mathcal{P}$ with $\delta(\mathcal{P}) < \delta_m$ satisfies

$$L(f, \mathcal{P}) > L(f, [a, b]) - \frac{1}{m} \quad \text{and} \quad U(f, \mathcal{P}) < U(f, [a, b]) + \frac{1}{m}.$$

Since $\delta(\mathcal{P}_n) \rightarrow 0$, for each m , there is an N_m such that $n > N_m$ implies $\delta(\mathcal{P}_n) < \delta_m$. Thus for each positive integer m , there exists an N_m such that $n > N_m$ implies

$$\begin{aligned} L(f, [a, b]) &\geq L(f, \mathcal{P}_n) > L(f, [a, b]) - \frac{1}{m} \iff |L(f, [a, b]) - L(f, \mathcal{P}_n)| < \frac{1}{m} \\ U(f, [a, b]) &\leq U(f, \mathcal{P}_n) < U(f, [a, b]) + \frac{1}{m} \iff |U(f, [a, b]) - U(f, \mathcal{P}_n)| < \frac{1}{m}. \end{aligned}$$

This is exactly what is meant by the statements

$$L(f, [a, b]) = \lim_{n \rightarrow \infty} L(f, \mathcal{P}_n) \quad \text{and} \quad U(f, [a, b]) = \lim_{n \rightarrow \infty} U(f, \mathcal{P}_n),$$

so we are done. □

Problem 3. Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \sum_{n=0}^{\infty} 2^{-n} \sin(2^n x)$$

is well-defined, bounded, and Riemann integrable on $[0, \pi]$. Find $\int_0^{\pi} f(x) dx$.

Solution. Since $f_n(x) = 2^{-n} \sin(2^n x)$ is bounded in absolute value by 2^{-n} for all real x and $\sum_{n=0}^{\infty} 2^{-n} = 2 < \infty$, the Weierstass M -test tells us that the sum $\sum_{n=0}^{\infty} f_n(x)$ converges uniformly on $\mathbb{R}$.

We can see that f is bounded because for any nonnegative integer N the triangle inequality lets us bound the partial sums

$$\left| \sum_{n=0}^N f_n(x) \right| \leq \sum_{n=0}^N |f_n(x)| \leq \sum_{n=0}^{\infty} 2^{-n} = 2$$

so that in the limit as N grows arbitrarily large we get

$$|f(x)| = \left| \sum_{n=0}^{\infty} f_n(x) \right| \leq 2.$$

Finally, since f is defined as a uniformly converging sum of continuous functions on $\mathbb{R}$, f is itself continuous on $\mathbb{R}$, hence Riemann integrable on $[0, \pi]$. Moreover, the Riemann integral commutes with uniform limits, so

$$\begin{aligned} \int_0^{\pi} f(x) dx &= \sum_{n=0}^{\infty} \int_0^{\pi} 2^{-n} \sin(2^n x) dx \\ &= \sum_{n=0}^{\infty} 4^{-n} (\cos(2^n \cdot 0) - \cos(2^n \cdot \pi)) \\ &= \cos(0) - \cos(\pi) + \sum_{n=1}^{\infty} 4^{-n} (\cos(0) - \cos(2^n \pi)) \\ &= 2 + \sum_{n=1}^{\infty} 4^{-n} (1 - 1) = \boxed{2} \end{aligned}$$

□

Problem 4. Axler's Definition 2.2 of outer measure on $\mathbb{R}$ uses *open* intervals. Show that the resulting outer measure would not change if *open* intervals were replaced by *closed*, or by *closed-open*, or by *open-closed*, or by *any*, i.e.

$$|A|_{()} = |A|_{[]} = |A|_{]]} = |A|_{[]} = |A|_{\text{any}} \quad \forall A \subseteq \mathbb{R}.$$

Solution. Given $A \subseteq \mathbb{R}$, define

$$\mathcal{L}_A^T = \left\{ \sum_{j=1}^{\infty} \ell(I_j) : I_1, I_2, \dots \text{ are bounded intervals of type } T \text{ such that } A \subseteq \bigcup_{j=1}^{\infty} I_j \right\},$$

where T indicates the status of the intervals' endpoints (open, closed, etc.); thus for example

$$|A|_{[]} = \inf \mathcal{L}_A^{[]} \quad \text{and} \quad |A|_{\text{any}} = \inf \mathcal{L}_A^{\text{any}}.$$

We stipulate that the intervals I_j are bounded because ∞ is trivially in each $\mathcal{L}_A^{\bullet}$ anyway, so we don't need to concern ourselves with intervals of infinite length.

Claim 4.1 — For any $A \subseteq \mathbb{R}$,

$$\mathcal{L}_A^{()} \subseteq \mathcal{L}_A^{\bullet} \subseteq \mathcal{L}_A^{\text{any}}.$$

Proof. Obviously $\mathcal{L}_A^{\text{any}}$ contains all the other $\mathcal{L}_A^{\bullet}$. For the other inclusion, we note that the open interval (a, b) is contained by each of $(a, b]$, $[a, b)$, and $[a, b]$, so if A is covered by a sequence of bounded intervals (a_1, b_1) , (a_2, b_2) , $\dots$, then it is covered by each of the sequences of intervals resulting from closing one or both of the endpoints of all the intervals (i.e. the sequences $(a_i, b_i]$, $[a_i, b_i)$, and $[a_i, b_i]$); therefore if $\sum_{j=1}^{\infty} (b_j - a_j)$ is in $\mathcal{L}_A^{()}$ then it is contained in the other $\mathcal{L}_A^{\bullet}$ as well. $\square$

Taking infima gives the inequality

$$|A|_{\text{any}} \leq |A|_{\bullet} \leq |A|_{()}$$

for all $A \subseteq \mathbb{R}$. To finish the solution, we will prove the reverse inequality

$$|A|_{()} \leq |A|_{\text{any}}.$$

Let $I_1, I_2, \dots$ be a sequence of (not necessarily open) intervals whose union contains A . Then, if I_j has endpoints $a_j < b_j$, we can see that $I_j \subseteq I'_j := (a_j - \varepsilon \cdot 2^{-j}, b_j + \varepsilon \cdot 2^{-j})$ for any $\varepsilon > 0$, so the I'_j are a covering of A by open intervals. Moreover,

$$\underbrace{\sum_{j=1}^{\infty} (b_j - a_j)}_{\text{total length of } I_j\text{'s}} \leq \underbrace{\sum_{j=1}^{\infty} (b_j - a_j + \varepsilon \cdot 2^{-j+1})}_{\text{total length of } I'_j\text{'s}} = \sum_{j=1}^{\infty} (b_j - a_j) + 2\varepsilon = \sum_{j=1}^{\infty} \ell(I_j) + 2\varepsilon$$

so $|A|_{()} \leq \sum_{j=1}^{\infty} \ell(I_j) + 2\varepsilon$ for all $\varepsilon > 0$, which implies that it is smaller than $\sum_{j=1}^{\infty} \ell(I_j)$. Taking the infimum over all choices of the I_j s gives $|A|_{()} \leq |A|_{\text{any}}$. $\square$

Problem 5. Show that the set

$$A = \left\{ x \in [-1, 1] : |\sin(nx)| \leq \frac{1}{n} \text{ for all } n \in \mathbb{N} \right\}$$

has (outer) measure zero.

Solution. Let $A_n = \{x \in [-1, 1] : |\sin(nx)| \leq \frac{1}{n}\}$, so $A = \bigcap_{n=1}^{\infty} A_n$. This means that $|A| \leq |A_n|$ for all n , so it suffices to prove that $\lim_{n \rightarrow \infty} |A_n| = 0$. To do this, we first rewrite A_n as

$$A_n = \left\{ x \in [-1, 1] : -\arcsin \frac{1}{n} \leq nx \bmod \pi \leq \arcsin \frac{1}{n} \right\}$$

where $nx \bmod \pi$ is the unique real number $r \in [0, \pi]$ for which there exists an integer q such that $nx = q\pi + r$. Equivalently,

$$A_n = \left\{ x \in [-1, 1] : \frac{-\arcsin \frac{1}{n}}{n} \leq x \bmod \frac{\pi}{n} \leq \frac{\arcsin \frac{1}{n}}{n} \right\}.$$

If $x \in [-1, 1]$ then $x + \frac{\pi}{n} \cdot n > 1$ and $x - \frac{\pi}{n} \cdot n < -1$, so each value of $x \bmod \frac{\pi}{n}$ corresponds to at most $2n$ points in A_n . This means that A_n is the union of up to $2n$ disjoint closed intervals, each with length at most $\frac{2 \arcsin \frac{1}{n}}{n}$, so

$$|A_n| \leq \frac{2 \arcsin \frac{1}{n}}{n} \cdot 2n$$

Taking the limit as n goes to infinity, we have

$$|A| \leq \lim_{n \rightarrow \infty} |A_n| \leq \lim_{n \rightarrow \infty} \frac{2 \arcsin \frac{1}{n}}{n} \cdot 2n = \lim_{n \rightarrow \infty} 4 \arcsin \frac{1}{n} = 4 \arcsin(0) = 0$$

using the fact that $\arcsin$ is continuous at 0, so $|A|$ is indeed zero. □

Problem 6. Let $A \subseteq [0, 1]$ be a subset of (outer) measure zero. Show that

$$B = \{x^2 : x \in A\}$$

is also of (outer) measure zero. Is the conclusion still true if $A \subseteq \mathbb{R}$?

Solution. We prove both parts via the following claim:

Claim 6.1 — Let A be a subset of the interval $[a, b]$ for which $0 \notin (a, b)$. If A has outer measure zero, so does $B = \{x^2 : x \in A\}$.

Proof. Let ε be a positive real number smaller than. We know by definition that there exists a sequence of open intervals $I_1, I_2, I_3, \dots$ whose union contains A such that $\sum_{j=1}^{\infty} \ell(I_j) < \frac{\varepsilon}{2 \max(|a|, |b|)}$. Define $I'_j = I_j \cap [a, b]$, so that each interval is contained in $[a, b]$, and let $I'_j = (a_j, b_j)$ for real numbers $a_j < b_j$. Then we have

$$\sum_{j=1}^{\infty} b_j - a_j = \sum_{j=1}^{\infty} \ell(I'_j) \leq \sum_{j=1}^{\infty} \ell(I_j) < \frac{\varepsilon}{2 \max(|a|, |b|)}.$$

Since $x \mapsto x^2$ is monotonic on $[a, b]$ (increasing if $0 \leq a < b$ and decreasing if $a < b \leq 0$), the set of intervals $\{(\min(a_j^2, b_j^2), \max(a_j^2, b_j^2)) : j \in \mathbb{N}\}$ covers B , and

$$\begin{aligned} \sum_{j=1}^{\infty} |b_j^2 - a_j^2| &= \sum_{j=1}^{\infty} (b_j - a_j) \cdot |b_j + a_j| \\ &\leq \sum_{j=1}^{\infty} (b_j - a_j) \cdot (|b_j| + |a_j|) \\ &\leq \sum_{j=1}^{\infty} (b_j - a_j) \cdot 2 \max(|a|, |b|) \\ &< 2 \max(|a|, |b|) \cdot \frac{\varepsilon}{2 \max(|a|, |b|)} = \varepsilon. \end{aligned}$$

Thus the outer measure of B is bounded above by ε for all ε , i.e. is zero. (Here we are using the result of Problem 4 so that some of the intervals in the covering I'_j of A may be half-open instead of open.) $\square$

This immediately solves the first part of the problem with $a = 0, b = 1$. For the second part of the problem, we can write $\mathbb{R} = \bigcup_{n \in \mathbb{Z}} [n, n+1]$, noting that 0 does not lie in the interior of any of these intervals. If $A \subseteq \mathbb{R}$ has measure zero, then by monotonicity so do its subsets $A \cap [n, n+1]$ for any $n \in \mathbb{Z}$; and then by the claim, $B_n = \{x^2 : x \in A \cap [n, n+1]\}$ also has outer measure zero. Obviously $B = \bigcup_{n \in \mathbb{Z}} B_n$, and the union of countably many sets of outer measure zero has outer measure zero (by countable subadditivity, since the outer measure of the union is bounded by the sum of countably many zeroes), which implies that $|B| = 0$. $\square$

Problem 7. Let C be the Cantor set and $\Lambda: [0, 1] \rightarrow [0, 1]$ the Cantor function.

(a) Suppose x is a rational number in $[0, 1]$. Explain why $\Lambda(x)$ is rational.

(b) Suppose $x \in C$ is such that $\Lambda(x)$ is rational. Explain why x is rational.

Solution. We will take for granted the fact that a real number $x \in [0, 1]$ is rational if and only if, for every integer $b \geq 2$, when x is expressed in base b as

$$x = \sum_{n=1}^{\infty} \frac{d_n}{b^n}$$

with digits $0 \leq d_n < b$, the sequence $(d_n)_{n \geq 1}$ is eventually periodic, which we say to mean that there exists an N such that $(d_n)_{n \geq N}$ is a periodic sequence.

(a) Let $x \in [0, 1]$ be rational. If $x \in C$, then

$$x \stackrel{\text{base } 3}{=} \sum_{n=1}^{\infty} \frac{d_n}{3^n} \implies \Lambda(x) \stackrel{\text{base } 2}{=} \sum_{n=1}^{\infty} \frac{d_n/2}{2^n}.$$

where the digits d_n of the base-3 expansion of x are in $\{0, 2\}$. Since $(d_n)_{n \geq 1}$ is eventually periodic (because x is rational), so is the sequence $(d_n/2)_{n \geq 1}$, so $\Lambda(x)$ is rational. On the other hand, if $x \notin C$, then let t be the smallest positive integer for which the base-3 expansion of x has a digit $d_t = 1$. Then,

$$\Lambda(x) = \sum_{n=1}^{t-1} \frac{d_n/2}{2^n}$$

which is rational because it is the sum of finitely many rational numbers.

(b) As stated in the solution to (a), if $x \in C$ then

$$x \stackrel{\text{base } 3}{=} \sum_{n=1}^{\infty} \frac{d_n}{3^n} \implies \Lambda(x) \stackrel{\text{base } 2}{=} \sum_{n=1}^{\infty} \frac{d_n/2}{2^n}.$$

Therefore if $\Lambda(x)$ is rational for some $x \in C$, the sequence $(d_n/2)_{n \geq 1}$ is eventually periodic, hence so is the sequence $(d_n)_{n \geq 1}$, implying that x is rational too. $\square$